

DAY 2

Diversity Analysis

Alpha diversity • Beta diversity • Ordination • Composition analysis • Statistics

Today's Learning Objectives

01 Alpha diversity

Calculate and interpret richness, evenness, and phylogenetic diversity metrics.

02 Beta diversity

Compute dissimilarity matrices (Bray-Curtis, UniFrac) to compare communities.

03 Ordination

Apply PCoA and NMDS to visualise community structure in reduced dimensions.

04 Composition analysis

Create taxonomy bar plots and heatmaps to show community composition.

05 Statistical testing

Use PERMANOVA, ANCOM-BC, and DESeq2 to identify significant differences.

06 R analysis

Implement diversity analyses using phyloseq, vegan, and ggplot2 in R.

Quick Recap: ASVs vs OTUs

We start from the feature table generated yesterday. Understanding ASVs vs OTUs is key to interpreting results:

Feature	OTUs (Operational Taxonomic Units)	ASVs (Amplicon Sequence Variants)
Definition	Sequences clustered at 97% similarity threshold	Exact sequences after error correction
Resolution	Genus level (typically)	Species/strain level possible
Reproducibility	Study-specific; cannot compare across studies easily	Identical ASVs in different studies = same sequence
Chimeras	Post-clustering removal (UCHIME)	Removed during error correction (DADA2)
Tools	UCLUST, VSEARCH, CD-HIT	DADA2, Deblur
Recommended?	Legacy; still used in some contexts	Yes — preferred modern approach

SECTION 1

Alpha Diversity

Richness • Evenness • Phylogenetic Diversity

Alpha Diversity: Within-Sample Diversity

Alpha diversity measures the diversity within a single sample.

Observed Features

Richness

Count of distinct ASVs/OTUs

Raw count of species — sensitive to sequencing depth. Must rarefy before comparing.

Chao1

Richness (estimated)

$S_{obs} + n_1^2 / (2n_2)$

Estimates total richness including unobserved rare species. n_1 = singletons, n_2 = doubletons.

Shannon Index (H')

Richness + Evenness

$H' = -\sum p_i \ln(p_i)$

Accounts for both species number AND how evenly distributed they are. Higher H = more diverse.

Simpson Index (1-D)

Evenness-weighted

$1 - \sum p_i^2$

Probability that two random individuals belong to different species. More weight on dominant species.

Faith's Phylogenetic Diversity

Phylogenetic

Sum of branch lengths in phylogeny

Captures evolutionary diversity — two communities with same richness may have very different PD.

Pielou Evenness (J')

Evenness

$J' = H' / \ln(S)$

Ranges 0–1. $J'=1$ means all species equally abundant. Tells us if one species dominates.

Rarefaction: Why Sequencing Depth Matters

Different samples may have different sequencing depths — this affects richness estimates. Rarefaction subsamples to equal depth for fair comparison.

The Problem

Sample A: 10,000 reads → 200 ASVs

Sample B: 50,000 reads → 350 ASVs

Is B really more diverse — or just sequenced deeper?

Rarefaction Curve

Plot species richness vs sequencing depth. Curve flattens when most diversity is captured. Used to choose rarefaction depth.

Rarefaction Depth

Choose depth where most samples plateau. Samples below threshold are excluded. qiime diversity alpha-rarefaction

Alternatives to Rarefaction

DESeq2 normalisation (size factors)

CLR transformation (ALDEx2)

Rarefaction loses data — debate continues.

QIIME2 command: `qiime diversity alpha-rarefaction --i-table table.qza --p-max-depth 10000`

SECTION 2

Beta Diversity

Bray-Curtis • UniFrac • Ordination

Beta Diversity: Between-Sample Diversity

Beta diversity measures how different communities are from each other. Computed as a dissimilarity matrix.

Bray-Curtis Dissimilarity

Abundance-based

Range: 0 (identical) $\rightarrow$ 1 (no overlap)

$$BC = \sum |A_i - B_i| / \sum (A_i + B_i)$$

Most widely used. Accounts for species presence AND relative abundance. Does not use phylogeny. Sensitive to dominant taxa.

Jaccard Distance

Presence/absence

Range: 0 $\rightarrow$ 1

$$J = 1 - |A \cap B| / |A \cup B|$$

Ignores abundance — only asks "is species present or not?". Sensitive to rare species. Good for sparse datasets.

Unweighted UniFrac

Phylogenetic (P/A)

Range: 0 $\rightarrow$ 1

Fraction of unique branch length

Uses phylogenetic tree. Presence/absence only. Detects differences in rare lineages not captured by abundance.

Weighted UniFrac

Phylogenetic (abundance)

Range: 0 $\rightarrow$ 1

Weighted by relative abundance

Phylogenetic AND abundance-weighted. Most comprehensive but may miss rare taxa. Recommended as primary metric.

Ordination: Visualising Community Structure

Ordination reduces the high-dimensional dissimilarity matrix to 2–3 dimensions for visualisation.

PCoA

Principal Coordinates Analysis

✓ Strengths

Preserves global structure well. Any distance matrix as input. Axes explain % variance.

✗ Limitations

Assumes linear relationships. Negative eigenvalues possible with some metrics.

Use when

Most common in microbiome. QIIME2 outputs PCoA by default (Emperor plots).

NMDS

Non-Metric Multidimensional Scaling

✓ Strengths

Handles non-linear relationships. Stress value measures goodness of fit. More flexible.

✗ Limitations

No variance explained %. Stochastic — results can vary between runs (set.seed!).

Use when

When PCoA gives poor representation. Commonly used with Bray-Curtis in vegan.

PCA

Principal Component Analysis

✓ Strengths

Standard in many fields. Loadings show which taxa drive variation. Fast computation.

✗ Limitations

Requires Euclidean distance. Not appropriate for raw counts (use CLR-transformed data).

Use when

Used with CLR-transformed data. ALDEx2 and ANCOM-BC workflows. Aitchison PCA.

Stress < 0.1 = good NMDS fit. PCoA axis labels show % variance explained. Always colour by metadata variable.

Taxonomy & Composition Analysis

Stacked Bar Plot

Most common. Shows relative abundance of each taxon per sample. Colour by phylum/genus. Group samples by metadata.

When: Overview of community composition across samples

QIIME2 taxa barplot or ggplot2 + phyloseq

Heatmap

Visualise abundance matrix (taxa × samples). Cluster rows/columns by hierarchical clustering. Log-transform abundance for better colour scale.

When: Compare top 20–50 taxa across many samples

pheatmap or ComplexHeatmap (R)

Bubble / Dot Plot

Plot prevalence vs mean abundance. Circle size = abundance, colour = taxon group. Good for identifying core vs. rare taxa.

When: Exploring prevalence-abundance relationships

ggplot2 + microbiome R package

Krona Chart

Interactive radial chart showing hierarchical taxonomy. Drill down from kingdom → phylum → genus. Interactive HTML output.

When: Visualising full taxonomic hierarchy interactively

KronaTools (ktImportTaxonomy)

Tip: Always group samples by a metadata variable. Show top 10–20 taxa; collapse the rest into "Other".

The Compositionality Problem

Microbiome data are compositional: we observe relative abundances (counts that sum to a fixed total), not absolute cell numbers. This violates assumptions of standard statistical tests.

Why a standard t-test or Wilcoxon FAILS on raw count data

Simplex constraint

All relative abundances sum to 1 (or 100%). If taxon A increases, others must proportionally decrease even if they did not change in absolute terms. This creates spurious negative correlations between taxa.

Subcompositional incoherence

The same taxon can appear to increase or decrease depending on which other taxa you include in the analysis. Your statistical result changes with filtering choices, not just with biology.

Sparsity and zero inflation

Most taxa have zero counts in many samples. Parametric tests assume continuous symmetric distributions. Zeros here are structural (true absences or below-detection limit), not missing data.

Beyond Compositionality: Absolute Abundance

Relative abundance only tells you proportions. To recover true biomass or copy-number changes, sequencing must be paired with an independent, absolute quantification method.

Three ways to escape the simplex — get real numbers, not proportions

Spike-in controls

Add a known quantity of a non-native DNA/cell standard (e.g. synthetic spike-in, mock community) before extraction. Back-calculate absolute abundance from the ratio of reads to spike-in signal.

qPCR / ddPCR total load

Quantify total 16S/bacterial copy number per gram or mL with qPCR or droplet digital PCR, then multiply relative abundances by this total to get per-taxon absolute abundance.

Flow cytometry cell counts

Directly count microbial cells per sample with flow cytometry (e.g. SYBR Green staining) — the gold standard for absolute abundance, independent of sequencing depth.

Compositional Differential Abundance Methods

ALDEx2

ANOVA-Like Differential Expression tool 2

Core idea: Centred Log-Ratio (CLR) Transformation

1. Monte Carlo Dirichlet sampling: generates many plausible count tables to model count uncertainty
2. CLR-transforms each table: $\log(x_i / \text{geometric mean})$ — removes the simplex constraint
3. Runs Welch t-test + Wilcoxon on CLR values across all Monte Carlo instances
4. Reports effect size (CLR difference) + expected p-value averaged over all instances

Best for 2-group comparisons. Conservative but robust FDR control. | Fernandes et al. 2014, PLOS Comp. Biol.

ANCOM-BC

Analysis of Compositions of Microbiomes with Bias Correction

Core idea: Log-Linear Model with Sampling Bias Correction

1. Models log-counts with a sample-specific bias term for unequal library sizes and technical variation
2. Estimates bias directly from data with no arbitrary reference taxon required
3. Tests each taxon with a linear model; supports covariates (age, BMI, batch) and multi-group designs
4. Controls FDR via Holm-Bonferroni or Benjamini-Hochberg correction across all taxa

Best for multi-group, longitudinal, or covariate-adjusted analyses. | Lin & Peddada 2020, Nature Communications.

SECTION 3

Statistical Testing

PERMANOVA • Differential Abundance • Effect Sizes

Community-Level Statistical Tests

PERMANOVA

`adonis2()` in *vegan*

Permutation-based multivariate ANOVA on distance matrix.
Tests whether group centroids differ (community structure).
Provides R^2 (effect size) and p-value. Can handle multiple variables.

```
QIIME2: qiime diversity adonis2  
R: adonis2(dist ~ group, data=meta)
```

ANOSIM

`anosim()` in *vegan*

Analysis of similarities.
Compares within-group vs between-group dissimilarities.
R statistic: 0=no difference, 1=perfect separation. Less powerful than PERMANOVA.

```
R: anosim(dist, grouping=meta$group)  
Interpret  $R > 0.25$  as meaningful separation
```

PERMDISP

`betadisper()` in *vegan*

Test for homogeneity of dispersion (variance).
Always run before PERMANOVA — PERMANOVA sensitive to unequal dispersion.
Tests whether groups differ in spread, not just centroid.

```
R: bd <- betadisper(dist, meta$group)  
permutest(bd) # test dispersion
```

Always check: n per group ≥ 3 | homogeneous dispersion | correct distance metric for your data type.

Differential Abundance Analysis

Identify which specific taxa (ASVs, genera) are significantly more/less abundant between groups.

ANCOM-BC

★ Recommended

Bias correction for sampling fraction. Provides p-values and effect sizes. QIIME2 q2-composition.

ANCOMBC R package | *qiime2 q2-composition*

DESeq2

Popular

Negative binomial model. Designed for RNA-seq but widely applied to microbiome. Log2 fold change with shrinkage.

Bioconductor: DESeq2

ALDEx2

Compositionally valid

Monte Carlo Dirichlet sampling + CLR transformation. Bayesian uncertainty. Fewer false positives.

Bioconductor: ALDEx2

LefSe

Classic (legacy)

LDA score for effect size. Widely used and cited but known for high false positive rate. Use with caution.

Galaxy web tool | *bioconda lefse*

Key concept: Microbiome data is compositional — always consider this when choosing a test.

SECTION 4

Microbial Co-occurrence Networks

SparCC • SPIEC-EASI • Network Topology

Microbial Co-occurrence Networks

SparCC

Sparse Correlations for Compositional data

Core idea: Iterative Log-Ratio Correlation Estimation

1. Estimates pairwise correlations between taxa using log-ratio variances, avoiding spurious correlations from standard Pearson/Spearman on compositional data
2. Iteratively refines estimates, excluding highly correlated pairs that bias the variance calculation
3. Uses bootstrapping/permutation to assign a p-value to each estimated correlation
4. Outputs a taxon-taxon correlation matrix used to build a co-occurrence network

Best for exploratory correlation networks from relative-abundance data. | Systematic evaluation: mSystems 2022, doi:10.1128/msystems.00961-22

SPIEC-EASI

SParse Inverse Covariance Estimation for Ecological Association Inference

Core idea: Graphical Model (Inverse Covariance) Estimation

1. CLR-transforms taxon abundances, then estimates a sparse inverse covariance (precision) matrix
2. Captures only direct associations — removes indirect correlations induced by shared confounders, unlike SparCC
3. Two estimation methods: Meinshausen-Bühlmann neighbourhood selection or graphical lasso
4. Network edges = non-zero entries in the precision matrix; visualise with igraph or NetCoMi

Best for direct, non-spurious association networks; sparser than SparCC. | Systematic evaluation: mSystems 2022, doi:10.1128/msystems.00961-22

R-based Diversity Analysis Workflow

01 Import to phyloseq

```
phyloseq(otu_table, sample_data,  
tax_table, phy_tree)
```

Combine feature table, metadata, taxonomy, and phylogeny into one phyloseq object.

02 Filter & Rarefy

```
prune_samples(sample_sums(ps)>1000, ps)  
rarefy_even_depth(ps, sample.size=5000)
```

Remove low-count samples, rarefy to equal depth, remove low-prevalence taxa.

03 Alpha Diversity

```
estimate_richness(ps, measures=  
c("Shannon", "Chao1", "Observed"))
```

Calculate alpha diversity metrics. Plot with ggplot2, test with Kruskal-Wallis or Wilcoxon.

04 Beta Diversity

```
dist <- phyloseq::distance(ps, "bray")  
ordinate(ps, "PCoA", distance=dist)
```

Compute distance matrix, run ordination, visualise with plot_ordination(). Run PERMANOVA.

05 Composition Plots

```
ps_genus <- tax_glom(ps, "Genus")  
plot_bar(ps_genus, fill="Phylum")
```

Agglomerate to genus/phylum level, create stacked bar plots, normalise to relative abundance.

06 Differential Abundance

```
library(ANCOMBC)  
ancombc2(data=ps, fix_formula="group")
```

Run ANCOM-BC2 or DESeq2, extract log2FC and adjusted p-values, plot volcano or forest plot.

Today's Hands-On Session

1 Alpha Diversity in R

Load QIIME2 output into phyloseq. Calculate Shannon, Chao1, Faith's PD. Plot boxplots by metadata group. Run Kruskal-Wallis test.

2 Beta Diversity & PCoA

Compute Bray-Curtis and weighted UniFrac. Run PCoA ordination. Plot Emperor (QIIME2) and ggplot2 (R) ordination plots. Run PERMANOVA.

3 Taxonomy Bar Plots

Generate stacked bar plots in QIIME2 and R (phyloseq + ggplot2). Agglomerate to phylum and genus. Normalise to relative abundance.

4 Differential Abundance

Run ANCOM-BC2 comparing two groups. Run DESeq2 on the same data. Compare results. Plot volcano plot of log2FC vs adjusted p-value.

Day 2 Summary

Alpha diversity

Measures within-sample diversity. Key metrics: Shannon (richness+evenness), Chao1 (estimated richness), Faith's PD (phylogenetic).

Rarefaction

Standardise sequencing depth before comparing alpha diversity. Choose depth at curve plateau.

Beta diversity

Between-sample dissimilarity. Bray-Curtis (abundance), UniFrac (phylogenetic). Stored as distance matrix.

Ordination

PCoA or NMDS reduces high-dimensional data to 2D for visualisation. Colour by metadata.

Statistics

PERMANOVA tests group-level differences. Always check dispersion (PERMDISP) first.

Differential abundance

ANCOM-BC or DESeq2 identify taxa differing between groups. Avoid naive t-tests on compositional data.

KEY REFERENCES

McMurdie & Holmes (2013)

phyloseq: R Package for Reproducible Microbiome Census Data Analysis

PLOS ONE 8: e61217

Oksanen et al. (2022)

vegan: Community Ecology Package (R)

R package v2.6-4 — CRAN

Lin & Peddada (2020)

Analysis of compositions of microbiomes with bias correction (ANCOM-BC)

Nature Communications 11: 3514

Love et al. (2014)

Moderated estimation of fold change and dispersion with DESeq2

Genome Biology 15: 550

Fernandes et al. (2014)

Unifying the analysis of high-throughput sequencing datasets (ALDEx2)

PLOS Computational Biology 10: e1003531

Faith (1992)

Conservation evaluation and phylogenetic diversity (Faith's PD)

Biological Conservation 61: 1–10